

Tutorial 1

Friday, February 7, 2025 05:36 PM

a. $\{w \mid w \text{ has at least three a's and at least two b's}\}$

→

$\{aaabb\}$

ababa

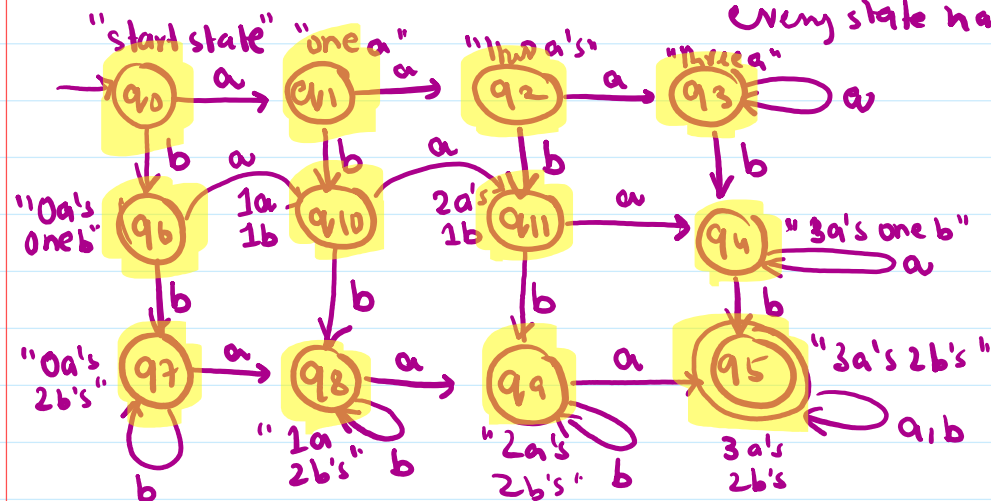
aaaaabbb ...

→ min length = 5

4 = aabbb x

$\Sigma = \{a, b\}$

every state has 2 transitions (DFA)



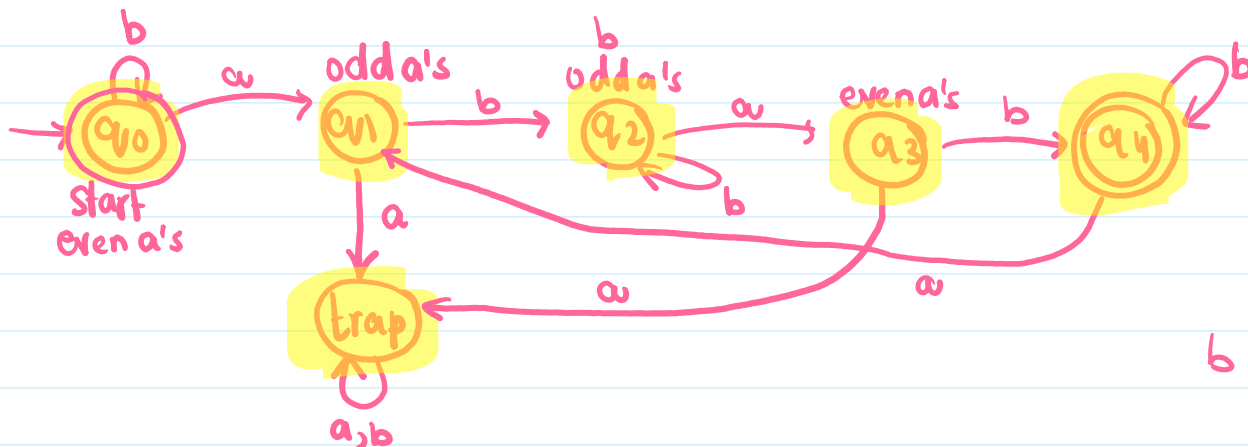
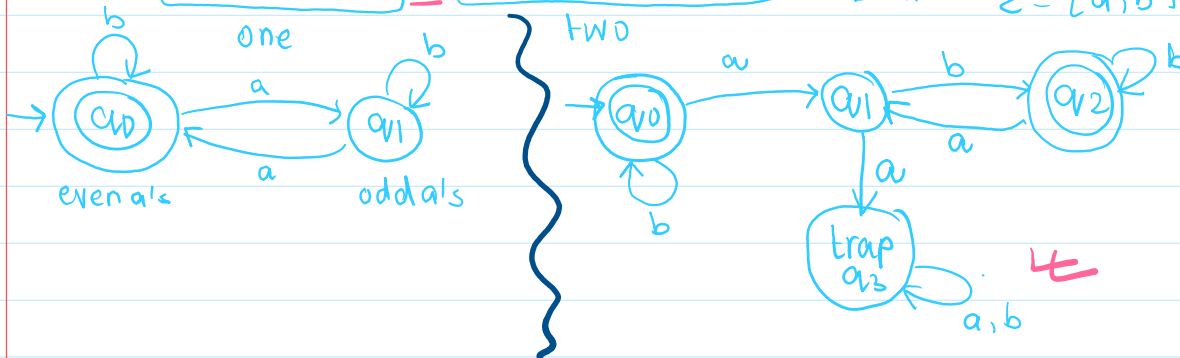
uaabbaaab

q4 state

d. $\{w \mid w \text{ has an even number of a's and each a is followed by at least one b}\}$

DFA

$\Sigma = \{a, b\}$



e. $\{w \mid w \text{ is any string not in } (ab^+)^*\}$

ab, abb, abbb, ...
a b a b

Whats not in it

ab, abb, abbb,

abab

(ab^*) repeating

Whats not in it



DFA - every state has

$\Sigma = \{ \}$ a transition of all characters from it

NFA - $n+1$ transitions up to

0, 1, 2, 3,

$\Sigma = \{a, b\}$

none, a, b, ϵ

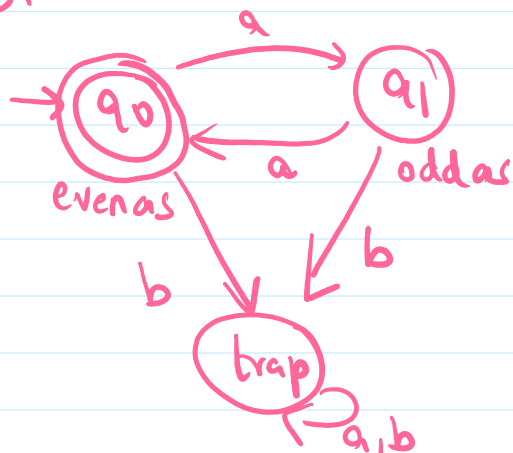
each string has even a's and no b's.

aaaa, aa, ϵ , aaaaaa,

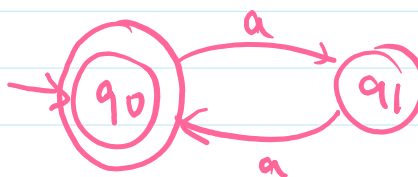
$(aa)^*$

$\Sigma = \{a, b\}$

DFA



NFA



e. $\{w \mid w \text{ is any string not in } (ab^+)^*\}$

DFA

$\downarrow$
+ at least 1
* 0 or more

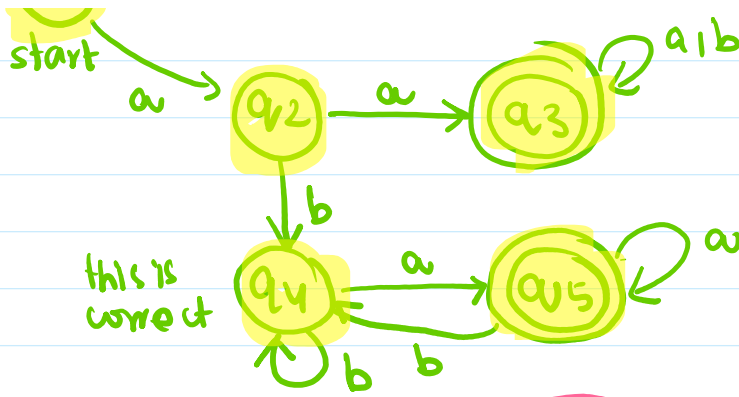
$(ab^+)^*$

$\begin{cases} b, \dots \\ a \checkmark \\ aa \checkmark \end{cases}$
aba, aab, ...



$(ab^+)^*$

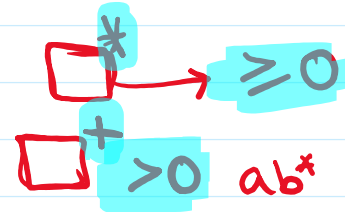
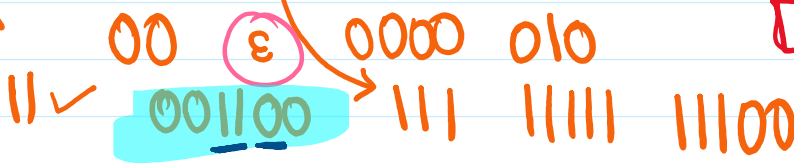
ba, ba...



baa
aba
abbbabab

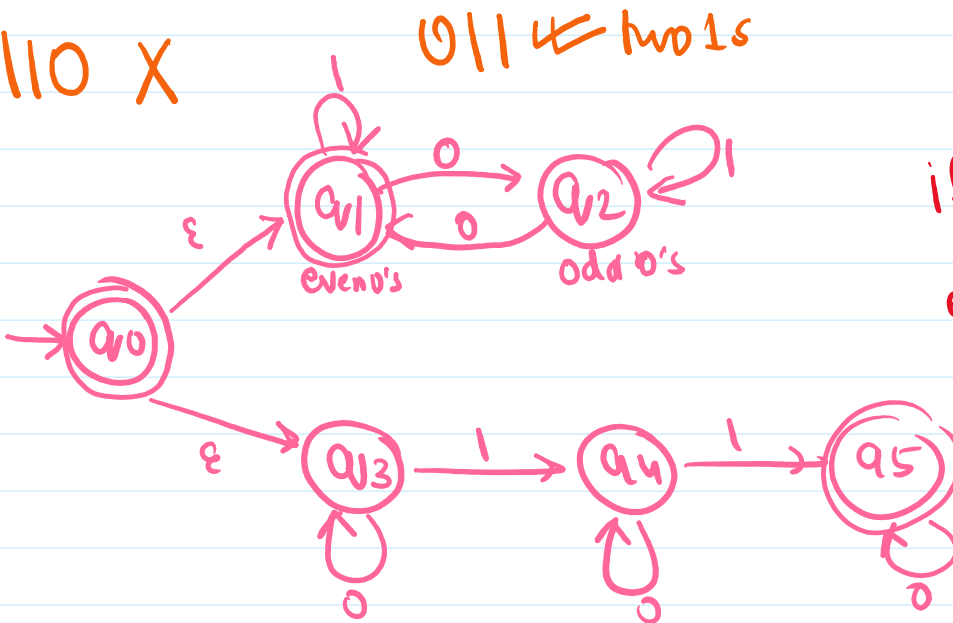
1. $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$

NFA



$a, \varepsilon = a$

NFA

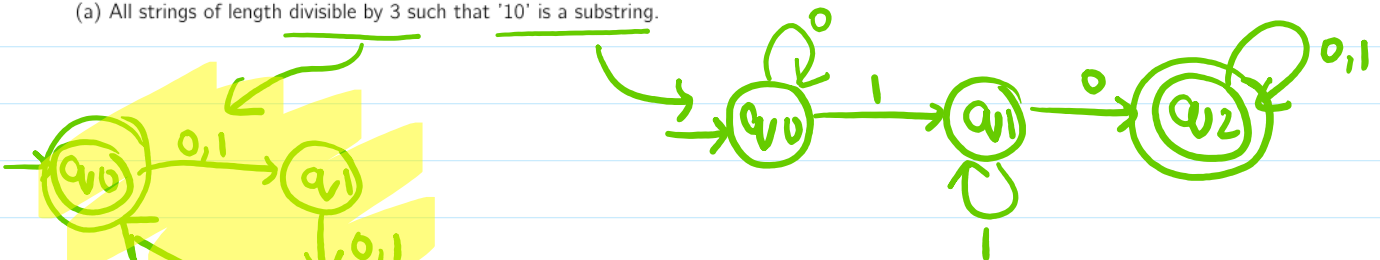


if (any copy = accept)
accept
else reject

for (for every copy)
if accept =
Overall ans
= accept
if overall ans = reject
reject

For each of the following languages L over the alphabet $\Sigma = \{0, 1\}$, give a regular expression that represents L and describe a DFA that recognizes it.

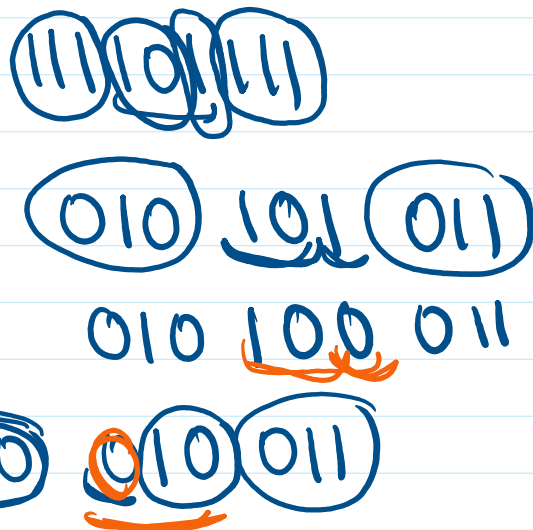
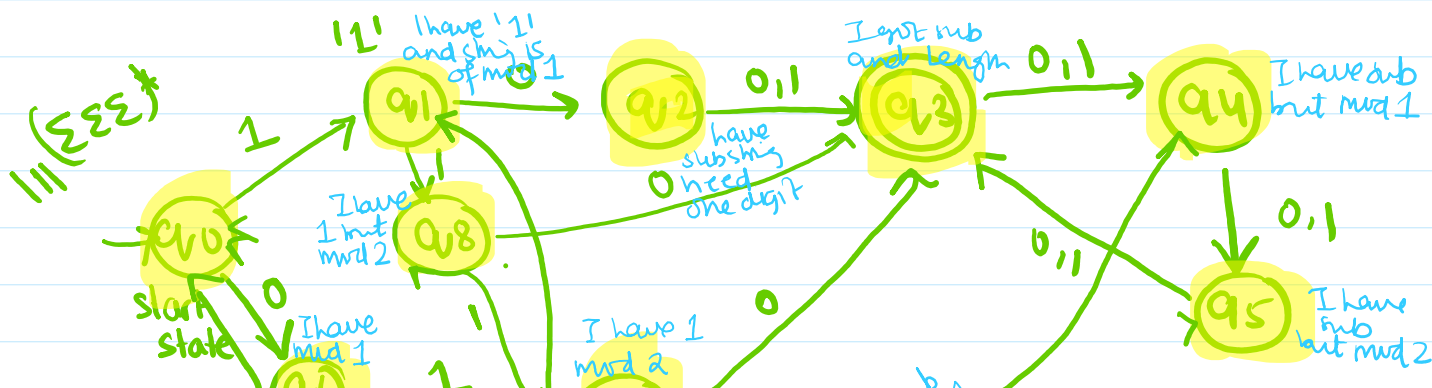
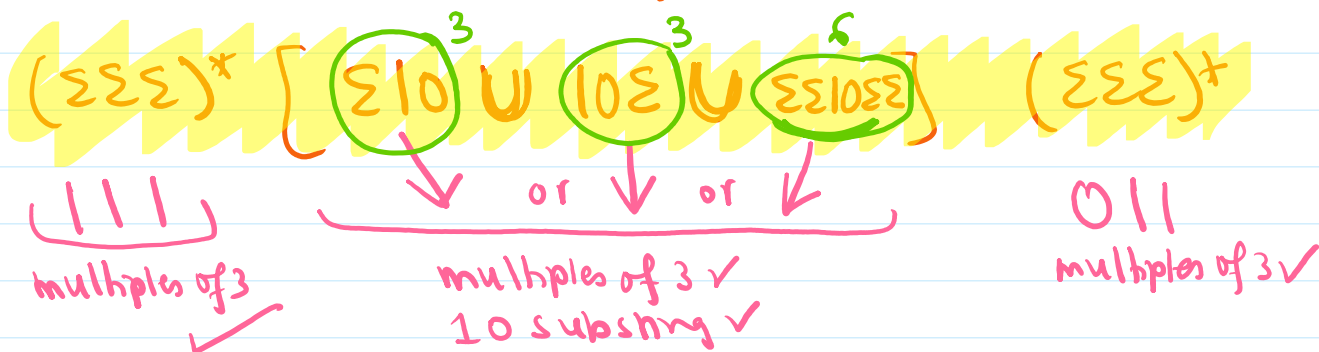
(a) All strings of length divisible by 3 such that '10' is a substring.




$$(333)^* \underbrace{10 \Sigma}_{\begin{smallmatrix} 101 \\ 100 \end{smallmatrix}} \neq (333)^*$$

② $(\Sigma\Sigma\Sigma)^* \quad \begin{matrix} \varepsilon & 10 \\ 1 & 10 \\ 0 & 10 \end{matrix} \quad (\Sigma\Sigma\Sigma)^*$

111011 $\rightarrow$ 4


$$\left. \begin{array}{l} \textcircled{2} \\ \left. \begin{array}{l} (\varepsilon \varepsilon \varepsilon)^* \quad \varepsilon 10 \quad (\varepsilon \varepsilon \varepsilon)^* \\ (333)^* \quad 013 \end{array} \right\} \\ \textcircled{1} \\ \left. \begin{array}{l} (\varepsilon \varepsilon \varepsilon)^* \quad 10\varepsilon \quad (\varepsilon \varepsilon \varepsilon)^* \\ (333)^* \quad 301 \end{array} \right\} \\ \textcircled{3} \\ \left. \begin{array}{l} (\varepsilon \varepsilon \varepsilon)^* \quad 3\varepsilon 10\varepsilon \varepsilon \quad (\varepsilon \varepsilon \varepsilon)^* \\ (333)^* \quad 330133 \end{array} \right\} \end{array} \right\}$$




(b) All strings that contain '101' as a substring followed by an even number of 1's.

$\Sigma = \{0, 1\}$, DFA

For each of the following languages L over the alphabet $\Sigma = \{0, 1\}$, construct an NFA with the specified number of states recognizing L .

(a) All strings have an odd number of 0's or 1's in multiples of 3, in six states.

(b) All strings are in the form $1^*0^*1^+$ with three states.